

# Retail Sales & Inventory Analysis

End-to-end retail sales and inventory analysis using Python, SQL, and Tableau to evaluate sales performance, profitability, supplier performance, and inventory risk.

NET SALES  
**\$10,482,110.25**

GROSS PROFIT  
**\$4,634,359.15**

GROSS MARGIN  
**44.21%**

INVENTORY AT COST  
**\$1,096,577.97**

Python

SQL

Tableau

# Business Context & Project Objective

Connecting retail sales performance with inventory risk to support better business decisions.

## Business questions answered

- ◆ Which categories, products, stores, and suppliers drive sales and gross profit?
- ◆ Where is gross margin strongest or weakest across the product mix?
- ◆ Which products create slow-moving, stockout, or overstock risk?
- ◆ How can inventory decisions improve revenue protection and working capital?

**Goal: turn raw retail sales and inventory files into reliable analytical outputs and clear business actions.**

## Project assets produced

1

### Clean dataset

Final analytical dataset with standardized sales, profit, and inventory measures

2

### Python notebook

Cleaning, transformation, feature creation, and export workflow

3

### SQL analysis

Validation and business questions using structured queries

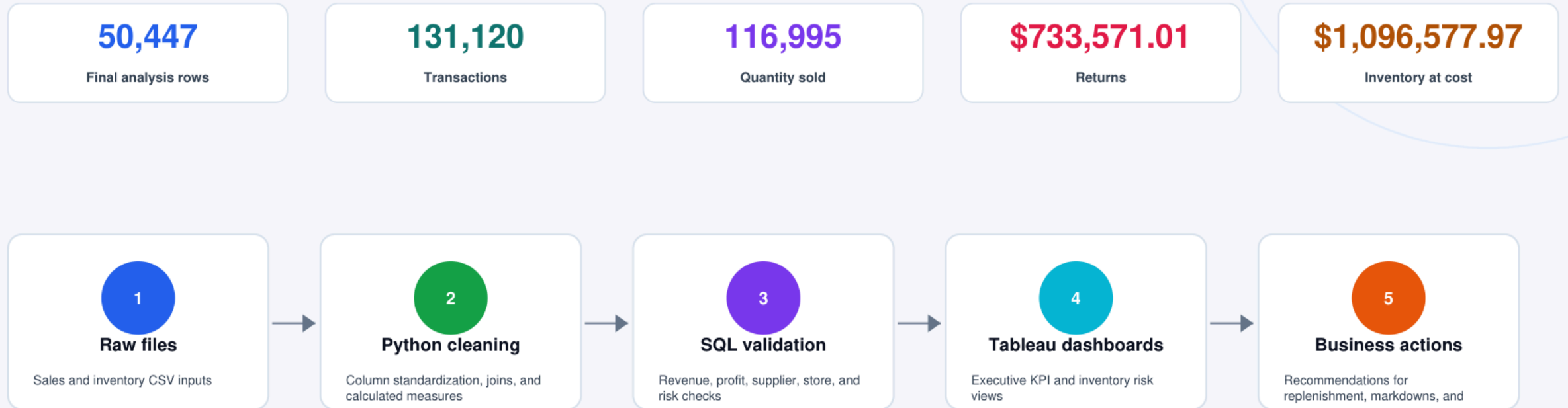
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### Tableau dashboards

Business Performance and Inventory Risk Analysis dashboards

# Analysis Workflow

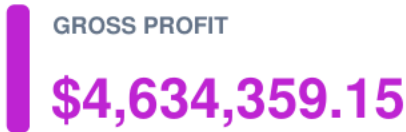
A reproducible pipeline from raw data to executive dashboards and recommendations.



**Quality control focus: KPI consistency across Python output, SQL checks, and Tableau dashboards to keep results credible and reproducible.**

# Executive KPI Summary

High-level retail performance and inventory position.



## Sales performance is strong

The business generated \$10.48M in net sales and \$4.63M in gross profit, supported by 131,120 transactions.



## Margin is a key strength

A 44.21% gross margin indicates meaningful profitability, but low-margin product groups still need review.



## Inventory requires active control

Current inventory at cost is \$1.10M, with slow-moving and overstock items creating working-capital pressure.

Sales, margin, product, store, and supplier performance in one executive view.

[illegible]

| Product Category      | Total Gross Profit (K) |
|-----------------------|------------------------|
| School Supplies       | 850                    |
| School Furniture      | 450                    |
| School Appliances     | 350                    |
| School Electronics    | 300                    |
| Furniture             | 250                    |
| Furniture Electronics | 150                    |
| School Electronics    | 100                    |
| Junior Electronics    | 50                     |
| Junior Furniture      | 40                     |
| School Electronics    | 30                     |
| Furniture             | 20                     |
| Junior Electronics    | 10                     |
| Junior Furniture      | 10                     |
| Junior Electronics    | 10                     |

| Product Category | Gross Margin% |
|------------------|---------------|
| Adult..          | 0.55          |
| School..         | 0.53          |
| Junior..         | 0.51          |
| School..         | 0.49          |
| Femm..           | 0.48          |
| School..         | 0.47          |
| School..         | 0.46          |
| Junior..         | 0.45          |
| Junior..         | 0.43          |
| School..         | 0.41          |
| School..         | 0.40          |
| School..         | 0.39          |
| School..         | 0.38          |
| Femm..           | 0.37          |
| Junior..         | 0.31          |

[illegible]

| Product Description | Total Gross Profit (K) |
|---------------------|------------------------|
| PROD_1              | 62                     |
| PROD_2              | 52                     |
| PROD_3              | 35                     |
| PROD_4              | 34                     |
| PROD_5              | 34                     |
| PROD_6              | 31                     |
| PROD_7              | 30                     |
| PROD_8              | 28                     |
| PROD_9              | 26                     |
| PROD_10             | 23                     |
| PROD_11             | 23                     |
| PROD_12             | 22                     |
| PROD_13             | 22                     |
| PROD_14             | 21                     |
| PROD_15             | 20                     |
| PROD_16             | 20                     |
| PROD_17             | 19                     |
| PROD_18             | 18                     |
| PROD_19             | 18                     |
| PROD_20             | 17                     |

| Product Description | Gross Margin% |
|---------------------|---------------|
| PROD_001            | 0.63          |
| PROD_002            | 0.63          |
| PROD_003            | 0.63          |
| PROD_004            | 0.63          |
| PROD_005            | 0.63          |
| PROD_006            | 0.63          |
| PROD_007            | 0.63          |
| PROD_008            | 0.63          |
| PROD_009            | 0.63          |
| PROD_010            | 0.63          |
| PROD_011            | 0.67          |
| PROD_012            | 0.67          |
| PROD_013            | 0.67          |
| PROD_014            | 0.67          |
| PROD_015            | 0.67          |
| PROD_016            | 0.67          |
| PROD_017            | 0.67          |
| PROD_018            | 0.67          |
| PROD_019            | 0.67          |
| PROD_020            | 0.67          |
| PROD_021            | 0.68          |

| Store    | Total Gross |         | Total Net Sa. |         |
|----------|-------------|---------|---------------|---------|
|          |             |         |               |         |
| STR-1006 | 485.56K     | 485.56K | 485.56K       | 485.56K |
| STR-1229 | 399.70K     | 399.70K | 399.70K       | 399.70K |
| STR-1090 | 371.28K     | 371.28K | 371.28K       | 371.28K |
| STR-1194 | 346.39K     | 346.39K | 346.39K       | 346.39K |
| STR-1010 | 343.44K     | 343.44K | 343.44K       | 343.44K |
| STR-1235 | 335.34K     | 335.34K | 335.34K       | 335.34K |
| STR-1148 | 332.40K     | 332.40K | 332.40K       | 332.40K |
| STR-1094 | 332.10K     | 332.10K | 332.10K       | 332.10K |
| STR-1222 | 328.17K     | 328.17K | 328.17K       | 328.17K |
| STR-1044 | 328.47K     | 328.47K | 328.47K       | 328.47K |
| STR-1075 | 327.93K     | 327.93K | 327.93K       | 327.93K |
| STR-1198 | 324.92K     | 324.92K | 324.92K       | 324.92K |
| STR-1253 | 303.17K     | 303.17K | 303.17K       | 303.17K |
| STR-1109 | 297.20K     | 297.20K | 297.20K       | 297.20K |
| STR-1203 | 284.13K     | 284.13K | 284.13K       | 284.13K |
| STR-1114 | 280.12K     | 280.12K | 280.12K       | 280.12K |
| STR-1107 | 274.41K     | 274.41K | 274.41K       | 274.41K |
| STR-1163 | 272.63K     | 272.63K | 272.63K       | 272.63K |
| STR-1157 | 270.73K     | 270.73K | 270.73K       | 270.73K |
| STR-1145 | 267.68K     | 267.68K | 267.68K       | 267.68K |
| STR-1038 | 258.64K     | 258.64K | 258.64K       | 258.64K |
| STR-1305 | 253.50K     | 253.50K | 253.50K       | 253.50K |
| STR-1350 | 253.08K     | 253.08K | 253.08K       | 253.08K |
| STR-1026 | 243.59K     | 243.59K | 243.59K       | 243.59K |
| STR-1074 | 241.74K     | 241.74K | 241.74K       | 241.74K |
| STR-1085 | 234.01K     | 234.01K | 234.01K       | 234.01K |
| STR-1338 | 232.05K     | 232.05K | 232.05K       | 232.05K |
| STR-1219 | 222.04K     | 222.04K | 222.04K       | 222.04K |
| STR-1092 | 217.61K     | 217.61K | 217.61K       | 217.61K |
| STR-1348 | 199.37K     | 199.37K | 199.37K       | 199.37K |
| STR-1255 | 193.64K     | 193.64K | 193.64K       | 193.64K |
| STR-1149 | 191.94K     | 191.94K | 191.94K       | 191.94K |
| STR-1237 | 191.89K     | 191.89K | 191.89K       | 191.89K |
| STR-1335 | 172.01K     | 172.01K | 172.01K       | 172.01K |
| STR-1353 | 157.29K     | 157.29K | 157.29K       | 157.29K |
| STR-1369 | 150.56K     | 150.56K | 150.56K       | 150.56K |
| STR-1234 | 150.44K     | 150.44K | 150.44K       | 150.44K |
| STR-1308 | 142.26K     | 142.26K | 142.26K       | 142.26K |
| STR-1105 | 139.45K     | 139.45K | 139.45K       | 139.45K |
| STR-1344 | 139.22K     | 139.22K | 139.22K       | 139.22K |

|                    |      | Supplier    |             |             |             |             |             |             |             |             |             |  |  |
|--------------------|------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--|--|
| Total Net Sales    | 3M   | 3.29M       | 3.20M       | 1.70M       | 0.54M       | 0.49M       | 0.42M       | 0.26M       | 0.18M       | 0.11M       | 0.08M       |  |  |
|                    | 1M   |             |             |             |             |             |             |             |             |             |             |  |  |
| Total Gross Profit | 1.5M | 3.29M       | 3.20M       | 1.70M       | 0.54M       | 0.49M       | 0.42M       | 0.26M       | 0.18M       | 0.11M       | 0.08M       |  |  |
|                    | 0.5M |             |             |             |             |             |             |             |             |             |             |  |  |
|                    |      | Vendor 0166 | Vendor 0134 | Vendor 0169 | Vendor 0246 | Vendor 0101 | Vendor 0064 | Vendor 0258 | Vendor 0106 | Vendor 0262 | Vendor 0054 |  |  |

Top suppliers dominate revenue

Category mix drives profit

● Returns need monitoring

Decision use: identify where to protect inventory, improve margin, and focus supplier conversations.

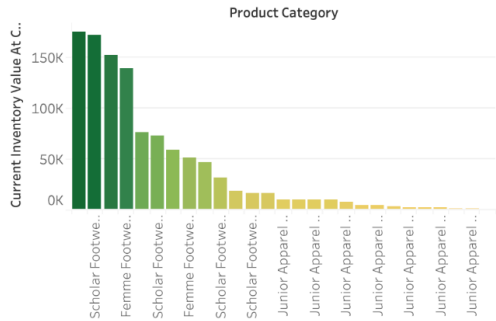
# Inventory Risk Analysis Dashboard

Inventory exposure, stockout risk, slow movement, and overstock pressure.

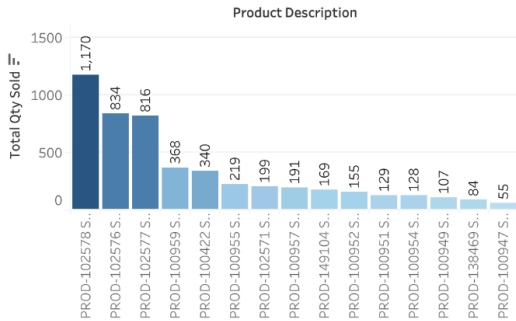
## Inventory Risk Summary

| Fast Moving Products | Slow Moving Products | Stockout Risk Products | Overstock Risk Products |
|----------------------|----------------------|------------------------|-------------------------|
| 13,997               | 21,658               | 10,323                 | 21,585                  |

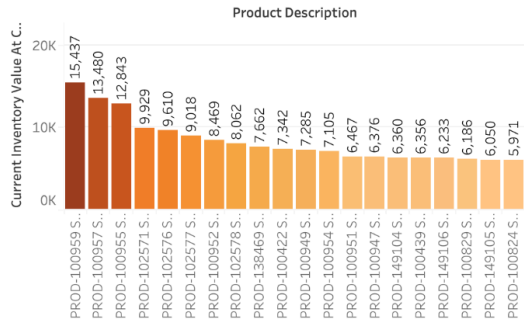
### Inventory Value



### Fast Moving Products



### Slow Moving Products



### Stockout Risk Products

|                                                                  |                                                               |                                         |                                 |  |  |
|------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------|---------------------------------|--|--|
| PROD-102578 SEG-500351 Scholar Footwear Flight Classic Zephyr    | PROD-102577 SEG-500351 Scholar Footwear Flight Classic Zephyr |                                         |                                 |  |  |
|                                                                  | PROD-102574 SEG-500351 Scholar Footwear Flight Classic Zephyr | PROD-102573 SEG-500351 Scholar Footwear |                                 |  |  |
| PROD-100439 SEG-500064 Scholar Footwear Boot Collection Woodland | PROD-100438 SEG-500064 Scholar Footwear Boot Collection       | PROD-100959 SEG-500123 Scholar Footwear |                                 |  |  |
| PROD-102576 SEG-500351 Scholar Footwear Flight Classic Zephyr    | PROD-102575 SEG-500351 Scholar Footwear Flight Classic        | PROD-101707 SEG-500224 Scholar Footwear | PROD-100422 SEG-500063 Scholar  |  |  |
|                                                                  |                                                               |                                         | PROD-100827 SEG-500112 Heritage |  |  |

### Overstock Risk Products

|                                                               |                                                               |                                                               |                                         |                                          |  |
|---------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------|------------------------------------------|--|
| PROD-100959 SEG-500123 Scholar Footwear Flight Classic Zephyr | PROD-102571 SEG-500351 Scholar Footwear Flight Classic Zephyr | PROD-102578 SEG-500351 Scholar Footwear Flight Classic Zephyr |                                         |                                          |  |
|                                                               | PROD-102576 SEG-500351 Scholar Footwear Flight Classic Zephyr | PROD-138469 SEG-508729 Scholar Footwear Modern Legend Summit  |                                         |                                          |  |
| PROD-100957 SEG-500123 Scholar Footwear Flight Classic Zephyr | PROD-102577 SEG-500351 Scholar Footwear Flight Classic Zephyr | PROD-100422 SEG-500063 Scholar Footwear Boot                  | PROD-149104 SEG-511422 Scholar Footwear |                                          |  |
| PROD-100955 SEG-500123 Scholar Footwear Flight Classic Zephyr | PROD-100952 SEG-500123 Scholar Footwear Flight Classic        | PROD-100949 SEG-500123 Scholar Footwear Flight                | PROD-149106 SEG-511422 Scholar Footwear | PROD-100824 SEG-500112 Heritage Footwear |  |

## Recommendations

Replenishment priority

Inventory control

Performance monitoring

Decision use: protect working capital while reducing stockout exposure in high-demand products.

# Executive Recommendations

Targeted actions across revenue growth, margin protection, and inventory risk control.



## Revenue Focus

Protect inventory for the highest revenue categories and top stores to prevent lost sales in proven demand areas.



## Margin Protection

Investigate low-margin products and supplier/category combinations before scaling promotions or purchasing.



## Inventory Control

Reduce exposure in slow-moving and overstock-risk products using markdown, transfer, or purchasing controls.



## Replenishment Priority

Monitor stockout-risk products more frequently and align restocking with sales velocity.



## Supplier Review

Use supplier rankings to validate dependency, negotiate performance terms, and improve product availability.



## Performance Monitoring

Track inventory, supplier, and sales KPIs monthly to support proactive decision making.

This project demonstrates data cleaning, SQL analysis, Tableau dashboarding, KPI validation, and business insight communication.